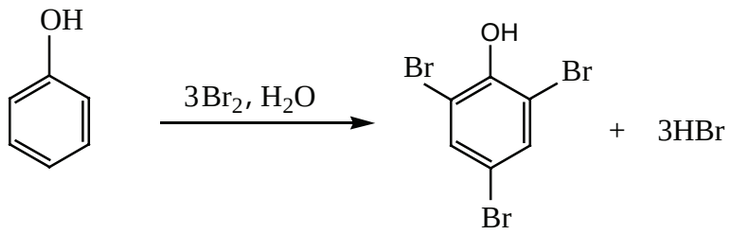
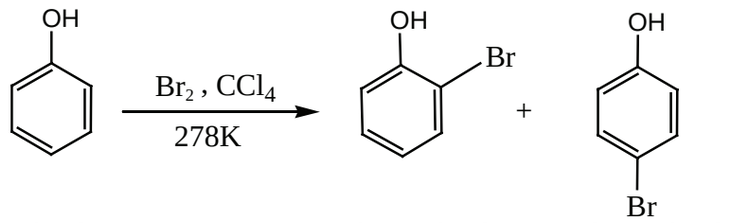
iv) Halogenation:

Phenol reacts with bromine water to give a white precipitate of 2,4,6-tri bromo phenol.



Phenol 2,4,6-tribromophenol

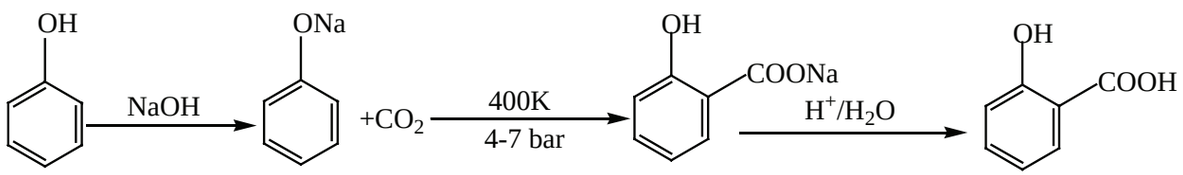
If the reaction is carried out in CS_2 or CCl_4 at 278K, a mixture of ortho and para bromo phenols are formed.



phenol o -bromo phenol p-bromo phenol

v) Kolbe's (or) Kolbe's Schmit reaction:

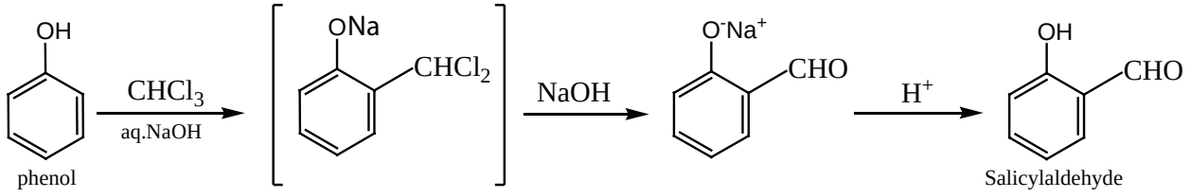
In this reaction, phenol is first converted into sodium phenoxide which is more reactive than phenol towards electrophilic substitution reaction with CO_2. Treatment of sodium phenoxide with CO_2 at 400K, 4-7 bar pressure followed by acid hydrolysis gives salicylic acid.



phenol sodium phenoxide sodium salicylate Salicyclic acid

vi) Riemer – Tiemann Reaction:

On treating phenol with $CHCl_3$/NaOH, a -CHO group is introduced at ortho position. This reaction proceeds through the formation of substituted benzal chloride intermediate.



phenol Salicylaldehyde

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